Covariance matrices and principal components

Linear Algebra · Lecture 9

Last time (Lecture 8): symmetric $A = Q\Lambda Q'$ (Q orthogonal, Λ diagonal); PSD $\iff$ eigenvalues $\geqslant 0$. Earlier results are cited where they are used; F1, F2: probability, without proof. $I = I_n$, $e_j = \mathrm{col}_j(I)$.

Facts on random variables (numbers whose values depend on chance; here with finite variances).

F1. $\mathrm{var}(u) = \mathsf{E}\Big((u-\mathsf{E}(u))^2\Big) \geqslant 0$, $\mathrm{cov}(u,v) = \mathsf{E}\Big((u-\mathsf{E}(u))(v-\mathsf{E}(v))\Big) = \mathrm{cov}(v,u)$, $\mathrm{var}(u) = \mathrm{cov}(u,u)$.
F2. $\mathsf{E}(au+bv+c) = a\,\mathsf{E}(u) + b\,\mathsf{E}(v) + c$ for numbers a, b, c (also for finite sums).

1. Random vectors and covariance matrices

Definition 1. A **random vector** $z = (z_1, \ldots, z_n)'$ or **random matrix** $Z = (z_{ij})$ has random variables as entries; $\mathsf{E}(Z) = (\mathsf{E}(z_{ij}))$. The **mean vector** of z is $\mu = \mathsf{E}(z)$, and its **covariance matrix** is

$$\Sigma = \mathrm{var}(z) = \mathsf{E}\Big((z-\mu)(z-\mu)'\Big) \in \mathbb{R}^{n\times n}$$

By Lecture 1, Theorem 4(d) and F1, $\sigma_{ij} = \mathsf{E}\Big((z_i-\mu_i)(z_j-\mu_j)\Big) = \mathrm{cov}(z_i, z_j)$, $\sigma_{ii} = \mathrm{var}(z_i)$, and Σ is symmetric.

Definition 2. For random vectors $u \in \mathbb{R}^p$, $v \in \mathbb{R}^q$ with $\mu_u = \mathsf{E}(u)$, $\mu_v = \mathsf{E}(v)$, the **covariance matrix of u and v** is

$$\mathrm{cov}(u,v) = \mathsf{E}\Big((u-\mu_u)(v-\mu_v)'\Big) \in \mathbb{R}^{p\times q}$$

Its (i,j) entry is $\mathrm{cov}(u_i, v_j)$. So $\mathrm{cov}(u,u) = \mathrm{var}(u)$ and $\mathrm{cov}(v,u) = \mathrm{cov}(u,v)'$ (Lecture 1, Theorem 8). u and v are **uncorrelated** if $\mathrm{cov}(u,v) = 0$.

Theorem 3. Let $z \in \mathbb{R}^n$ be a random vector, $\mathsf{E}(z) = \mu$, $\mathrm{var}(z) = \Sigma$.

(a) $\mathsf{E}(AZB + D) = A\,\mathsf{E}(Z)B + D$ for a random $Z \in \mathbb{R}^{m\times p}$ and constant $A \in \mathbb{R}^{q\times m}$, $B \in \mathbb{R}^{p\times l}$, $D \in \mathbb{R}^{q\times l}$.

(b) $\mathsf{E}(Az+b) = A\mu + b$ and $\mathrm{var}(Az+b) = A\Sigma A'$ for constant $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$.

(c) $a'\Sigma a = \mathrm{var}(a'z) \geqslant 0$ for $a \in \mathbb{R}^n$: Σ is PSD.

Proof. (a) Entry (i,j) of $AZB + D$ is $\sum_r \sum_s a_{ir} z_{rs} b_{sj} + d_{ij}$ (Lecture 1, Theorem 4(d)); its expectation is $\sum_r \sum_s a_{ir}\,\mathsf{E}(z_{rs}) b_{sj} + d_{ij}$ (F2), entry (i,j) of $A\,\mathsf{E}(Z)B + D$.

(b):

$$\mathsf{E}(Az+b) = A\mu + b, \qquad Az + b - \mathsf{E}(Az+b) = A(z-\mu) \qquad \text{((a): } Z = z,\ B = 1,\ D = b)$$
$$\mathrm{var}(Az+b) = \mathsf{E}\Big(A(z-\mu)(z-\mu)'A'\Big) = A\Sigma A' \qquad \text{(Lecture 1, Theorem 8; (a))}$$

(c):

$$\mathrm{var}(a'z) = a'\Sigma a \geqslant 0 \qquad \text{((b) with } a',\ b = 0;\ \text{F1)}$$

Σ is symmetric (Definition 1): PSD (Lecture 8, Definition 13: $x'\Sigma x \geqslant 0$ for all x).

□

Theorem 4. Let $u \in \mathbb{R}^p$, $v \in \mathbb{R}^q$ be random vectors, $\mu_u = \mathsf{E}(u)$, $\mu_v = \mathsf{E}(v)$.

(a) $\operatorname{cov}(Au + b, Bv + d) = A \operatorname{cov}(u, v) B'$ for constant $A \in \mathbb{R}^{k\times p}$, $b \in \mathbb{R}^k$, $B \in \mathbb{R}^{l\times q}$, $d \in \mathbb{R}^l$. So $\operatorname{cov}(Az, Bz) = A\Sigma B'$ if $\operatorname{var}(z) = \Sigma$.

(b) $\operatorname{cov}(u_1+u_2, v) = \operatorname{cov}(u_1, v)+\operatorname{cov}(u_2, v)$ and $\operatorname{cov}(v, u_1+u_2) = \operatorname{cov}(v, u_1)+\operatorname{cov}(v, u_2)$ $(u_1, u_2 \in \mathbb{R}^p)$.

(c) If $x_1, \ldots, x_m \in \mathbb{R}^p$ and $\operatorname{cov}(x_i, x_j) = 0$ for $i \neq j$, then $\operatorname{var}(x_1+\cdots+x_m) = \operatorname{var}(x_1)+\cdots+\operatorname{var}(x_m)$.

(d) $\mathsf{E}(uu') = \operatorname{var}(u) + \mu_u\mu_u'$.

Proof.

(a) By Theorem 3(b): $Au + b - \mathsf{E}(Au + b) = A(u - \mu_u)$, $Bv + d - \mathsf{E}(Bv + d) = B(v - \mu_v)$; transposes (Lecture 1, Theorem 8):

$$\operatorname{cov}(Au + b, Bv + d) = \mathsf{E}\Big(A(u - \mu_u)(v - \mu_v)'B'\Big) = A \operatorname{cov}(u, v) B' \qquad \text{(Theorem 3(a))}$$

$u = v = z$: $\operatorname{cov}(Az, Bz) = A \operatorname{var}(z) B'$.

(b) $\mathsf{E}(u_1 + u_2) = \mathsf{E}(u_1) + \mathsf{E}(u_2)$, so $(u_1 + u_2 - \mathsf{E}(u_1 + u_2))(v - \mu_v)' = (u_1 - \mathsf{E}(u_1))(v - \mu_v)' + (u_2 - \mathsf{E}(u_2))(v - \mu_v)'$ (Lecture 1, Theorem 3(a)); take expectations entry by entry (F2). Transpose for the second identity (Definition 2).

(c) By (b) in both arguments, $\operatorname{var}(\sum_i x_i) = \operatorname{cov}(\sum_i x_i, \sum_j x_j) = \sum_i \sum_j \operatorname{cov}(x_i, x_j)$; only the m terms with $i = j$ remain.

(d) $(u - \mu_u)(u - \mu_u)' = uu' - u\mu_u' - \mu_u u' + \mu_u\mu_u'$ (Lecture 1, Theorems 3(a), 8); take expectations entry by entry, with $\mathsf{E}(u\mu_u') = \mu_u\mu_u' = \mathsf{E}(\mu_u u')$ (Theorem 3(a)): $\operatorname{var}(u) = \mathsf{E}(uu') - \mu_u\mu_u'$. □

Example 5. Let $z \in \mathbb{R}^4$ be a random vector with

$$\operatorname{var}(z) = \Sigma = \begin{pmatrix} 4 & 0 & 4 & 0 \\ 0 & 1 & 2 & 2 \\ 4 & 2 & 9 & 6 \\ 0 & 2 & 6 & 9 \end{pmatrix}$$

Find $\operatorname{cov}(z, e)$ and $\operatorname{var}(e)$ for $e = z_3 - z_1 - 2z_2$ and for $f = z_4 + 2z_1 + 2z_2 - 2z_3$; $\operatorname{var}(z_1 + z_2 + z_3 + z_4)$; the variances and the covariance of $z_1 + z_2$ and $z_1 - z_2$.

Solution.

$e = a'z$, $a = (-1, -2, 1, 0)'$; $f = b'z$, $b = (2, 2, -2, 1)'$; by Theorem 4(a), $\operatorname{cov}(z, a'z) = \Sigma a$, $\operatorname{var}(a'z) = a'\Sigma a$:

$$\Sigma a = (-4 + 0 + 4 + 0,\ 0 - 2 + 2 + 0,\ -4 - 4 + 9 + 0,\ 0 - 4 + 6 + 0)' = (0, 0, 1, 2)'$$
$$\Sigma b = (8 + 0 - 8 + 0,\ 0 + 2 - 4 + 2,\ 8 + 4 - 18 + 6,\ 0 + 4 - 12 + 9)' = (0, 0, 0, 1)'$$

So $\operatorname{cov}(z, e) = (0, 0, 1, 2)'$ and $\operatorname{var}(e) = a'\Sigma a = 0 + 0 + 1 + 0 = 1$: e is uncorrelated with z_1, z_2; $\operatorname{cov}(z, f) = (0, 0, 0, 1)'$ and $\operatorname{var}(f) = b'\Sigma b = 1$: f is uncorrelated with z_1, z_2, z_3.

$z_1 + z_2 + z_3 + z_4 = \mathbf{1}'z$, $z_1 + z_2$, $z_1 - z_2$ (Theorem 4(a)):

$$\operatorname{var}(\mathbf{1}'z) = \mathbf{1}'\Sigma\mathbf{1} = 51 \qquad \text{(the sum of all entries of } \Sigma)$$
$$(1, 1, 0, 0)\Sigma = (4, 1, 6, 2), \qquad (1, -1, 0, 0)\Sigma = (4, -1, 2, -2)$$

So $\operatorname{var}(z_1+z_2) = 4+1 = 5$, $\operatorname{var}(z_1-z_2) = 4+1 = 5$, and $\operatorname{cov}(z_1+z_2, z_1-z_2) = (4, 1, 6, 2)(1, -1, 0, 0)' = 4 - 1 = 3$.

Theorem 6. Let $z \in \mathbb{R}^n$ be a random vector, $\mathsf{E}(z) = \mu$, $\mathrm{var}(z) = \Sigma$, and $A \in \mathbb{R}^{n\times n}$ constant. Then $\mathsf{E}(z'Az) = \mathrm{tr}(A\Sigma) + \mu' A\mu$; so $\mathsf{E}(z'Az) = \mathrm{tr}(A\Sigma)$ if $\mu = 0$.

Proof.

$x'Ax = \sum_i \sum_j a_{ij}x_i x_j$ for $x = z$ and $x = \mu$ (Lecture 8, Definition 12); Σ is symmetric (Definition 1); $\mathrm{tr}(A\Sigma) = \sum_i \sum_j a_{ij}\sigma_{ji}$ (Lecture 1, Definition 9, Theorem 4(d)):

$$\begin{aligned}
\mathsf{E}(z'Az) &= \textstyle\sum_i \sum_j a_{ij}\, \mathsf{E}(z_i z_j) && \text{(F2)}\\
\sigma_{ij} &= \mathsf{E}\Big((z_i - \mu_i)(z_j - \mu_j)\Big) = \mathsf{E}(z_i z_j) - \mu_i\mu_j - \mu_j\mu_i + \mu_i\mu_j && \text{(F1, F2)}\\
\mathsf{E}(z'Az) &= \textstyle\sum_i \sum_j a_{ij}\sigma_{ij} + \sum_i \sum_j a_{ij}\mu_i\mu_j = \mathrm{tr}(A\Sigma) + \mu' A\mu && (\sigma_{ji} = \sigma_{ij})
\end{aligned}$$

□

Definition 7. Let observations be modeled as a random vector y whose distribution depends on unknown parameters. An **estimator** of a parameter θ is a random variable $T = g(y)$, where the function g does not involve unknown parameters; T is **unbiased** if $\mathsf{E}(T) = \theta$ for all values of the parameters.

Example 8. Let $y \in \mathbb{R}^n$, $\mathsf{E}(y) = \mu$, $\mathrm{var}(y) = \sigma^2 I_n$, and $P \in \mathbb{R}^{n\times n}$ with $P' = P = P^2$ and $\mathrm{rank}\, P = r$. Find $\mathsf{E}(y'Py)$, and the expectation of $s^2 = \frac{1}{n-1}\sum_i (y_i - \bar{y})^2$.

Solution. $\mathrm{tr}\, P = \mathrm{rank}\, P = r$ (Lecture 8, Theorem 11: symmetric $A^2 = A \Rightarrow \mathrm{rank}\, A = \mathrm{tr}\, A$), and by Theorem 6 and Lecture 1, Theorem 10(a), $\mathsf{E}(y'Py) = \mathrm{tr}(P\sigma^2 I_n) + \mu' P\mu = \sigma^2 r + \mu' P\mu$.

- $P = C_n = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}'$ ($n \geqslant 2$; Lecture 1, Definition 14): $C_n' = C_n = C_n^2$, $C_n\mathbf{1} = 0$ (Lecture 1, Proposition 15), $\mathrm{rank}\, C_n = n - 1$ (Lecture 8, Example 5); $y'C_n y = \sum_i (y_i - \bar{y})^2$ and $\mu' C_n \mu = \sum_i (\mu_i - \bar{\mu})^2$ ($\bar{y} = \frac{1}{n}\mathbf{1}'y$; Lecture 1, Example 16). So $\mathsf{E}\Big(\sum_i (y_i - \bar{y})^2\Big) = (n-1)\sigma^2 + \sum_i (\mu_i - \bar{\mu})^2$.
- Common mean, $\mu = \mu_0\mathbf{1}$: $C_n\mu = 0$, and $\mathsf{E}(s^2) = \sigma^2$ for all μ_0, σ^2: s^2 is an unbiased estimator of σ^2 (Definition 7). The estimator $\frac{1}{n}\sum_i (y_i - \bar{y})^2$ has the expectation $\frac{n-1}{n}\sigma^2 < \sigma^2$ ($\sigma^2 > 0$): not unbiased.
- Drifting means: for $n = 4$ and $\mu = (1, 3, 5, 7)'$, $\bar{\mu} = 4$ and $\sum_i (\mu_i - \bar{\mu})^2 = 9 + 1 + 1 + 9 = 20$, so $\mathsf{E}(s^2) = \sigma^2 + \frac{20}{3}$: s^2 overestimates σ^2 on average.

2. Data matrices and the sample covariance matrix

Definition 9. A **data matrix** is $X = (x_{ij}) \in \mathbb{R}^{n\times p}$ ($n \geqslant 2$): x_{ij} is the value of variable j on unit i; row i of X is $x_i' = (x_{i1}, \ldots, x_{ip})$. With $\mathbf{1} = \mathbf{1}_n$ and $C_n = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}'$ (Lecture 1, Definition 14): the **mean vector** $\bar{x} = \frac{1}{n}X'\mathbf{1} = (\bar{x}_1, \ldots, \bar{x}_p)'$, $\bar{x}_j = \frac{1}{n}\sum_i x_{ij}$; the **centered data matrix** $C_n X = X - \mathbf{1}\bar{x}'$ (Lecture 1, Example 17), with entries $x_{ij} - \bar{x}_j$; the **sample covariance matrix** $S = (s_{jl}) = \frac{1}{n-1}X'C_n X \in \mathbb{R}^{p\times p}$. For $y, w \in \mathbb{R}^n$, $\frac{1}{n-1}\sum_i (y_i - \bar{y})(w_i - \bar{w})$ is their **sample covariance** (y, w are **uncorrelated** if it is 0), and $\frac{1}{n-1}\sum_i (y_i - \bar{y})^2$ is the **sample variance** of y. For $a \in \mathbb{R}^p$, $Xa = (a'x_1, \ldots, a'x_n)'$ holds the values on the n units of the linear combination of the variables with coefficient vector a.

Proposition 10. Let $X \in \mathbb{R}^{n\times p}$ be a data matrix, with $\bar{x}$ and S as in Definition 9.

(a) $X'C_nX = (C_nX)'(C_nX) = X'X - n\bar{x}\bar{x}'$; s_{jj} is the sample variance of variable j and s_{jl} the sample covariance of variables j and l; S is symmetric and PSD.

(b) For $a, b \in \mathbb{R}^p$: Xa has mean $a'\bar{x}$ and centered values $C_n(Xa) = (C_nX)a$; the sample covariance of Xa and Xb is $a'Sb$, and the sample variance of Xa is $a'Sa$.

Proof.

(a) $C_n' = C_n = C_n^2$ (Lecture 1, Proposition 15), $X'\mathbf{1} = n\bar{x}$; column j of C_nX holds $x_{ij} - \bar{x}_j$:

$$\begin{aligned} X'C_nX &= X'C_n'C_nX = (C_nX)'(C_nX) && \text{(Lecture 1, Theorem 8)} \\ X'C_nX &= X'X - \tfrac{1}{n}(X'\mathbf{1})(\mathbf{1}'X) = X'X - n\bar{x}\bar{x}' && \text{(Lecture 1, Theorems 3(a), 8)} \\ \Big((C_nX)'(C_nX)\Big)_{jl} &= \textstyle\sum_i (x_{ij} - \bar{x}_j)(x_{il} - \bar{x}_l) && \text{(Lecture 1, Theorem 4(d))} \end{aligned}$$

Divide by $n-1$: s_{jl}, s_{jj}. S is symmetric (Lecture 1, Theorem 8) and PSD (Lecture 8, Theorem 15(b): $A'A$ is PSD).

(b) $\frac{1}{n}\mathbf{1}'(Xa) = \bar{x}'a = a'\bar{x}$, $C_n(Xa) = (C_nX)a$: the deviations of Xa from its mean (Lecture 1, Theorem 8, Example 16):

$$\tfrac{1}{n-1}(C_nXa)'(C_nXb) = a'\Big(\tfrac{1}{n-1}(C_nX)'(C_nX)\Big)b = a'Sb \qquad \text{(Lecture 1, Theorem 8; (a))}$$

$b = a$: the sample variance of Xa is $a'Sa$.

□

Example 11. Five units ($n = 5$) are measured on four variables ($p = 4$), for instance five students on four tests:

$$X = \begin{pmatrix} 7 & 3 & 7 & 3 \\ 9 & 9 & 5 & 7 \\ 5 & 7 & 9 & 9 \\ 3 & 5 & 1 & 1 \\ 1 & 1 & 3 & 5 \end{pmatrix}$$

Find $\bar{x}$, C_5X and S.

Solution.

$\mathbf{1}'X$, $\bar{x}$; subtract $\bar{x}'$ from every row (Definition 9):

$$\mathbf{1}'X = (25,\ 25,\ 25,\ 25), \qquad \bar{x} = \tfrac{1}{5}X'\mathbf{1} = \begin{pmatrix} 5 \\ 5 \\ 5 \\ 5 \end{pmatrix}, \qquad C_5X = \begin{pmatrix} 2 & -2 & 2 & -2 \\ 4 & 4 & 0 & 2 \\ 0 & 2 & 4 & 4 \\ -2 & 0 & -4 & -4 \\ -4 & -4 & -2 & 0 \end{pmatrix}$$

$(C_5X)'(C_5X)$: inner products of the columns (Lecture 1, Theorem 4(d)); S (Proposition 10(a)):

$$(C_5X)'(C_5X) = \begin{pmatrix} 40 & 28 & 20 & 12 \\ 28 & 40 & 12 & 20 \\ 20 & 12 & 40 & 28 \\ 12 & 20 & 28 & 40 \end{pmatrix}, \qquad S = \frac{1}{4}(C_5X)'(C_5X) = \begin{pmatrix} 10 & 7 & 5 & 3 \\ 7 & 10 & 3 & 5 \\ 5 & 3 & 10 & 7 \\ 3 & 5 & 7 & 10 \end{pmatrix}$$

For instance $40 = 4 + 16 + 0 + 4 + 16$ (column 1) and $28 = -4 + 16 + 0 + 0 + 16$ (columns 1, 2). Every variable has the sample variance 10; the pairs $\{1,2\}$, $\{3,4\}$ have the sample covariance 7, the pairs $\{1,3\}$, $\{2,4\}$ have 5, and $\{1,4\}$, $\{2,3\}$ have 3.

Theorem 12. Let $x_1, \dots, x_n \in \mathbb{R}^p$ ($n \geqslant 2$) be random vectors with $\mathsf{E}(x_i) = \mu$, $\operatorname{var}(x_i) = \Sigma$ and $\operatorname{cov}(x_i, x_j) = 0$ for $i \neq j$; X the random data matrix with rows $x_1', \dots, x_n'$, $\bar{x} = \frac{1}{n}X'\mathbf{1}$ and $S = \frac{1}{n-1}X'C_nX$.

(a) $\mathsf{E}(\bar{x}) = \mu$ and $\operatorname{var}(\bar{x}) = \frac{1}{n}\Sigma$.

(b) $\mathsf{E}(S) = \Sigma$.

Proof. (a) $X'\mathbf{1} = x_1 + \cdots + x_n$ (Lecture 1, Theorem 4(b)), so $\mathsf{E}(\bar{x}) = \frac{1}{n}(\mu + \cdots + \mu) = \mu$ (Theorem 3(a)), and $\operatorname{var}(\bar{x}) = \frac{1}{n^2}\operatorname{var}(x_1 + \cdots + x_n) = \frac{1}{n^2}n\Sigma = \frac{1}{n}\Sigma$ (Theorem 3(b) with $A = \frac{1}{n}I_p$, Theorem 4(c)).
(b) $\mathsf{E}(x_ix_i') = \Sigma + \mu\mu'$ and $\mathsf{E}(\bar{x}\bar{x}') = \frac{1}{n}\Sigma + \mu\mu'$ (Theorem 4(d); (a)). By Proposition 10(a) and Lecture 1, Theorem 4(d), $X'C_nX = X'X - n\bar{x}\bar{x}'$ with $X'X = x_1x_1' + \cdots + x_nx_n'$; so $\mathsf{E}(X'C_nX) = n(\Sigma + \mu\mu') - n(\frac{1}{n}\Sigma + \mu\mu') = (n-1)\Sigma$, and $\mathsf{E}(S) = \Sigma$. □

With the divisor n in place of $n-1$, the expectation would be $\frac{n-1}{n}\Sigma$, not Σ (if $\Sigma \neq 0$); this is why S has the divisor $n-1$.

Definition 13. Let $X \in \mathbb{R}^{n\times p}$ be a data matrix with every $s_{jj} > 0$; $\sqrt{s_{jj}}$ is the **sample standard deviation** of variable j. With $D_s = \operatorname{diag}(s_{11}, \dots, s_{pp})$ and $D_s^{-1/2} = \operatorname{diag}\Big(1/\sqrt{s_{11}}, \dots, 1/\sqrt{s_{pp}}\Big)$, $r_{jl} = s_{jl}/\sqrt{s_{jj}s_{ll}}$ is the **sample correlation** of variables j and l, and $R = (r_{jl}) = D_s^{-1/2}SD_s^{-1/2}$ (Lecture 1, Proposition 13(a)) is the **sample correlation matrix**.

Proposition 14. In Definition 13, $r_{jj} = 1$ and $|r_{jl}| \leqslant 1$.

Proof.
$u_j = \operatorname{col}_j(C_nX)$: $(n-1)s_{jl} = u_j'u_l$ (Proposition 10(a)), $\|u_j\|^2 = u_j'u_j$:

$$r_{jj} = \frac{s_{jj}}{s_{jj}} = 1, \qquad r_{jl} = \frac{u_j'u_l}{\|u_j\|\,\|u_l\|}, \qquad |r_{jl}| \leqslant 1 \qquad \text{(Lecture 6, Theorem 3: } |x'y| \leqslant \|x\|\,\|y\|\text{)}$$

□

Example 11: every $s_{jj} = 10$, so $D_s^{-1/2} = \frac{1}{\sqrt{10}}I_4$ and $R = \frac{1}{10}S$:
$r_{12} = s_{12}/\sqrt{s_{11}s_{22}} = 7/\sqrt{10 \cdot 10} = \frac{7}{10}$, $r_{13} = \frac{5}{10} = \frac{1}{2}$, $r_{14} = \frac{3}{10}$.

3. The Rayleigh quotient

Theorem 15. Let $A \in \mathbb{R}^{n\times n}$ be symmetric, and let $q_1, \dots, q_n$ be orthonormal (Lecture 6, Definition 8) eigenvectors of A, $Aq_i = \lambda_iq_i$, numbered so that $\lambda_1 \geqslant \cdots \geqslant \lambda_n$ (Lecture 8, Theorem 9(a)); then $Q = [q_1 \ \cdots \ q_n]$ is orthogonal, $Q'AQ = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, and the λ_i are the eigenvalues of A (Lecture 8, Theorem 9(b)). For $x \neq 0$, $x'Ax/x'x$ is the **Rayleigh quotient** of x.

(a) $\lambda_n \leqslant x'Ax/x'x \leqslant \lambda_1$ for every $x \neq 0$, with equality at $x = q_n$ and $x = q_1$: $\max_{x\neq 0} x'Ax/x'x = \lambda_1$ and $\min_{x\neq 0} x'Ax/x'x = \lambda_n$.

(b) For $x \neq 0$: $x'Ax = \lambda_1x'x \iff Ax = \lambda_1x$, and $x'Ax = \lambda_nx'x \iff Ax = \lambda_nx$.

Proof.
(a) $y = Q'x$: $x = QQ'x = Qy$ ($QQ' = I$, Lecture 6, Theorem 9(b)):

$$\begin{aligned} x'x = y'Q'Qy = y'y, \qquad x'Ax = \lambda_1y_1^2 + \cdots + \lambda_ny_n^2 \qquad & \text{(Lecture 8, Theorem 14)} \\ \lambda_nx'x = \textstyle\sum_i \lambda_ny_i^2 \leqslant \sum_i \lambda_iy_i^2 \leqslant \sum_i \lambda_1y_i^2 = \lambda_1x'x \qquad & (\lambda_n \leqslant \lambda_i \leqslant \lambda_1) \\ q_1'Aq_1 = \lambda_1q_1'q_1 = \lambda_1, \qquad q_n'Aq_n = \lambda_nq_n'q_n = \lambda_n & \end{aligned}$$

Divide by $x'x > 0$.

(b) $x \neq 0$, $y = Q'x$, $x = Qy = \sum_i y_i q_i$ (Lecture 1, Theorem 4(b)):

$$\lambda_1 x'x - x'Ax = \sum_i (\lambda_1 - \lambda_i) y_i^2 \geqslant 0$$
$$\lambda_1 x'x - x'Ax = 0 \iff y_i = 0 \text{ for } \lambda_i < \lambda_1 \implies x = \sum_{\lambda_i = \lambda_1} y_i q_i \implies Ax = \lambda_1 x$$
$$Ax = \lambda_1 x \implies x'Ax = \lambda_1 x'x$$

For λ_n, apply this to $-A$: $(-A)q_i = -\lambda_i q_i$, and in the order $q_n, \ldots, q_1$ the eigenvalues $-\lambda_n \geqslant \cdots \geqslant -\lambda_1$ of $-A$ decrease.

□

Theorem 16. Let A, $q_1, \ldots, q_n$ and $\lambda_1 \geqslant \cdots \geqslant \lambda_n$ be as in Theorem 15, and $1 \leqslant k \leqslant n$. Then

$$\lambda_k = \max\left\{ \frac{x'Ax}{x'x} : x \neq 0,\ x \perp q_1, \ldots, q_{k-1} \right\}$$

attained at $x = q_k$ (for $k = 1$ there is no condition); $x \perp q$ means $q'x = 0$ (Lecture 6, Definition 1).

Proof.

$Q = [R\ T]$, $R = [q_1\ \cdots\ q_{k-1}]$, $T = [q_k\ \cdots\ q_n]$; $QQ' = I$ (Lecture 6, Theorem 9(b)); $AT = [\lambda_k q_k\ \cdots\ \lambda_n q_n] = T\,\mathrm{diag}(\lambda_k, \ldots, \lambda_n)$ (Lecture 1, Theorem 4(a), Proposition 13(a)):

$$R'T = 0,\ T'T = I_{n-k+1},\ I = QQ' = RR' + TT' \quad \text{(Lecture 1, Theorem 4(d))}$$
$$x \perp q_1, \ldots, q_{k-1} \iff R'x = 0 \iff x = Tz,\ z = T'x \quad \text{(Lecture 1, Theorem 4(c); line 1)}$$
$$x'x = z'T'Tz = z'z, \qquad x'Ax = z'(T'AT)z$$
$$T'AT = T'T\,\mathrm{diag}(\lambda_k, \ldots, \lambda_n) = \mathrm{diag}(\lambda_k, \ldots, \lambda_n)$$

By Theorem 15(a) for $\mathrm{diag}(\lambda_k, \ldots, \lambda_n)$ (with $Q = I$), the max of $z'(T'AT)z/z'z$ over $z \neq 0$ is λ_k, at $z = e_1$: $x = Te_1 = q_k$.

□

4. Principal components

Definition 17. Let $X \in \mathbb{R}^{n \times p}$ be a data matrix with $S \neq 0$. A **first loading vector** is a unit vector $v_1 \in \mathbb{R}^p$ that maximizes $a'Sa$ over all unit vectors $a \in \mathbb{R}^p$ ($\|a\| = 1$, Lecture 6, Definition 1). For $m \geqslant 2$, once $v_1, \ldots, v_{m-1}$ are chosen, an m**th loading vector** is a unit vector v_m that maximizes $a'Sa$ over the unit vectors a with $a'Sv_l = 0$ ($l = 1, \ldots, m-1$) (the m**th problem**). The linear combination with coefficient vector v_m is the m**th principal component**, and $z_m = C_n X v_m \in \mathbb{R}^n$, with the entries $z_{im} = v_m'(x_i - \bar{x})$, is the m**th score vector**.

By Proposition 10(b), $a'Sa$ is the sample variance of Xa, and $a'Sv_l = 0$ says that Xa and Xv_l are uncorrelated: the mth principal component has the largest sample variance among the combinations with a unit coefficient vector that are uncorrelated with the first $m-1$ principal components.

Theorem 18. Let $X \in \mathbb{R}^{n \times p}$ be a data matrix with $S \neq 0$, and $S = V\Lambda V'$ with $V = [v_1\ \cdots\ v_p]$ orthogonal, $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_p)$, $\lambda_1 \geqslant \cdots \geqslant \lambda_p$ (as in Theorem 15). Let $r = \mathrm{rank}(C_n X)$ and $z_m = C_n X v_m$.

(a) $\lambda_p \geqslant 0$, $r \leqslant \min\{n-1, p\}$, and $\lambda_m > 0 \iff m \leqslant r$.

(b) If $\lambda_l > 0$, then for every $a \in \mathbb{R}^p$: Xa and Xv_l are uncorrelated $\iff a \perp v_l$.

(c) Let $m \leqslant r$. If $v_1, \ldots, v_{m-1}$ are the first $m-1$ loading vectors, then v_m is an mth loading vector, and the maximum in the mth problem is λ_m.

(d) $z_m' z_l = (n-1)\lambda_m$ if $l = m$, and 0 if $l \neq m$: the score vectors are pairwise orthogonal and uncorrelated, and z_m has the sample variance λ_m.

Proof. (a)

- S is PSD (Proposition 10(a)), so $\lambda_p \geqslant 0$ (Lecture 8, Theorem 14(a)).
- $r = \operatorname{rank}(C_nX) \leqslant \operatorname{rank} C_n = n-1$ (Lecture 4, Theorem 15(a); Lecture 8, Example 5), $r \leqslant p$ (Lecture 4, Theorem 12(a)).
- $\operatorname{rank} S = \operatorname{rank}\Big((C_nX)'(C_nX)\Big) = r$ (Lecture 4, Theorems 15(d), 16(b)) and $\operatorname{rank} S = \#\{m : \lambda_m \neq 0\}$ (Lecture 8, Theorem 9(b)). As $\lambda_1 \geqslant \cdots \geqslant \lambda_p \geqslant 0$, exactly $\lambda_1, \ldots, \lambda_r$ are positive.

(b) By Proposition 10(b) and $Sv_l = \lambda_l v_l$, the sample covariance of Xa and Xv_l is $a'Sv_l = \lambda_l a'v_l$, which is $0 \iff a'v_l = 0$, as $\lambda_l > 0$.

(c) Unit a, $c = V'a$: $a = Vc$, $c_j = v_j'a$, $\sum_j c_j^2 = \|c\|^2 = \|a\|^2 = 1$ (Lecture 6, Theorem 9(a),(b)); by (b), (a), the mth problem asks $c_1 = \cdots = c_{m-1} = 0$:

$$\begin{gathered} a'Sa = c'V'SVc = c'\Lambda c = \textstyle\sum_j \lambda_j c_j^2 \\ \lambda_m - a'Sa = \textstyle\sum_{j\geqslant m} \lambda_m c_j^2 - \sum_{j\geqslant m} \lambda_j c_j^2 = \sum_{j>m} (\lambda_m - \lambda_j) c_j^2 \geqslant 0 \\ v_m'Sv_l = \lambda_l v_m'v_l = 0 \ (l < m), \qquad v_m'Sv_m = \lambda_m \end{gathered}$$

So v_m solves the mth problem, with the maximum λ_m.

(d) Proposition 10(a); $\frac{1}{n}\mathbf{1}'z_m = \frac{1}{n}\mathbf{1}'C_nXv_m = 0$ (Lecture 1, Proposition 15):

$$z_m'z_l = v_m'(C_nX)'(C_nX)v_l = (n-1)v_m'Sv_l = (n-1)\lambda_l v_m'v_l$$

The score vectors have mean 0, so $\frac{1}{n-1}z_m'z_l$ is their sample covariance (Definition 9).

□

Part (c) is Theorem 16 for S.

Proposition 19. In Theorem 18, the **total variance** is $s_{11} + \cdots + s_{pp} = \operatorname{tr} S = \lambda_1 + \cdots + \lambda_p$, and the **proportion of variance explained** by the mth principal component is

$$\mathrm{PVE}_m = \frac{\|z_m\|^2}{\|C_nX\|_F^2} = \frac{\lambda_m}{\lambda_1 + \cdots + \lambda_p} \qquad (m = 1, \ldots, p)$$

where $\|C_nX\|_F^2 = \operatorname{tr}\Big((C_nX)'(C_nX)\Big)$ is the sum of the squared entries of C_nX (Lecture 1, Theorem 10(c)). $\mathrm{PVE}_m > 0$ for $m \leqslant r$, $\mathrm{PVE}_m = 0$ for $m > r$, $\mathrm{PVE}_1 \geqslant \mathrm{PVE}_2 \geqslant \cdots$, and $\mathrm{PVE}_1 + \cdots + \mathrm{PVE}_M$ increases with $M \leqslant r$ and equals 1 for $M \geqslant r$.

Proof.

$V'V = I$ (Lecture 6, Definition 8); Proposition 10(a), Theorem 18(d):

$$\begin{aligned} \operatorname{tr} S = \operatorname{tr}(V\Lambda V') = \operatorname{tr}(\Lambda V'V) = \operatorname{tr}\Lambda = \lambda_1 + \cdots + \lambda_p \qquad & \text{(Lecture 1, Theorem 10(b))} \\ \|C_nX\|_F^2 = \operatorname{tr}\Big((C_nX)'(C_nX)\Big) = (n-1)\operatorname{tr} S, \qquad \|z_m\|^2 = (n-1)\lambda_m & \end{aligned}$$

$\operatorname{tr} S \geqslant \lambda_1 > 0$ (Theorem 18(a); $\Lambda \neq 0$ as $S = V\Lambda V' \neq 0$); $\lambda_m > 0$ exactly for $m \leqslant r$, and λ_m decreases: $\mathrm{PVE}_1 + \cdots + \mathrm{PVE}_r = (\lambda_1 + \cdots + \lambda_r)/\operatorname{tr} S = 1$.

□

Example 20. Find the principal components, the score vectors and the proportions of variance explained for the data of Example 11.

Solution. $h_1 = \mathbf{1}_4$, $h_2 = (1,1,-1,-1)'$, $h_3 = (1,-1,1,-1)'$, $h_4 = (1,-1,-1,1)'$.

Sh_m row by row (S of Example 11); the h_m are orthogonal, $\|h_m\| = 2$:

$$Sh_1 = \begin{pmatrix}25\\25\\25\\25\end{pmatrix} = 25h_1, \quad Sh_2 = \begin{pmatrix}9\\9\\-9\\-9\end{pmatrix} = 9h_2, \quad Sh_3 = \begin{pmatrix}5\\-5\\5\\-5\end{pmatrix} = 5h_3, \quad Sh_4 = \begin{pmatrix}1\\-1\\-1\\1\end{pmatrix} = h_4$$

For instance $10+7+5+3 = 25$, $10+7-5-3 = 9$, $10-7+5-3 = 5$, $10-7-5+3 = 1$ in row 1. So $v_m = \frac{1}{2}h_m$ are orthonormal eigenvectors, the eigenvalues are $\lambda_1 = 25$, $\lambda_2 = 9$, $\lambda_3 = 5$, $\lambda_4 = 1$, and $S = V\,\mathrm{diag}(25,9,5,1)V'$ with $V = [v_1 \ \cdots \ v_4]$ (Lecture 8, Theorem 9(b)); $r = 4$ (Theorem 18(a)).

By Theorem 18(c), the first principal component, with coefficient vector $\frac{1}{2}(1,1,1,1)'$ and sample variance 25, is half the sum of the four variables: an overall level. The second, $\frac{1}{2}(1,1,-1,-1)'$, with sample variance 9, contrasts variables 1, 2 with variables 3, 4; the third, $\frac{1}{2}(1,-1,1,-1)'$, with sample variance 5, contrasts variables 1, 3 with 2, 4; the fourth, $\frac{1}{2}(1,-1,-1,1)'$, has sample variance 1.

Scores $z_m = C_5Xv_m = \frac{1}{2}C_5Xh_m$ (C_5X of Example 11); check (Theorem 18(d)):

$$C_5Xh_1 = \begin{pmatrix}0\\10\\10\\-10\\-10\end{pmatrix}, \quad C_5Xh_2 = \begin{pmatrix}0\\6\\-6\\6\\-6\end{pmatrix}, \quad C_5Xh_3 = \begin{pmatrix}8\\-2\\-2\\-2\\-2\end{pmatrix}, \quad C_5Xh_4 = \begin{pmatrix}0\\2\\-2\\-2\\2\end{pmatrix}$$

$$z_1 = \begin{pmatrix}0\\5\\5\\-5\\-5\end{pmatrix}, \quad z_2 = \begin{pmatrix}0\\3\\-3\\3\\-3\end{pmatrix}, \quad z_3 = \begin{pmatrix}4\\-1\\-1\\-1\\-1\end{pmatrix}, \quad z_4 = \begin{pmatrix}0\\1\\-1\\-1\\1\end{pmatrix}$$

$\|z_1\|^2 = 100 = 4\cdot 25$, $\|z_2\|^2 = 36 = 4\cdot 9$, $\|z_3\|^2 = 20 = 4\cdot 5$, $\|z_4\|^2 = 4 = 4\cdot 1$, and $z_m'z_l = 0$ for $m < l$; for instance $z_1'z_2 = 0+15-15-15+15 = 0$ and $z_1'z_3 = 0-5-5+5+5 = 0$.

Proportions (Proposition 19):

$\mathrm{tr}\,S = 10+10+10+10 = 40 = 25+9+5+1$, so $\mathrm{PVE}_1 = \frac{25}{40} = \frac{5}{8}$, $\mathrm{PVE}_2 = \frac{9}{40}$, $\mathrm{PVE}_3 = \frac{5}{40} = \frac{1}{8}$, $\mathrm{PVE}_4 = \frac{1}{40}$: the first component explains $\frac{5}{8}$ of the total variance, the first two together $\frac{34}{40} = \frac{17}{20}$.

Remark 21. Usually only the first M principal components are kept, with M much smaller than p; their score vectors $z_1, \ldots, z_M$ summarize the data, for instance in a plot of the units. By Proposition 19, they explain together the proportion $\mathrm{PVE}_1 + \cdots + \mathrm{PVE}_M$ of the total variance, and $\mathrm{PVE}_1 \geqslant \mathrm{PVE}_2 \geqslant \cdots$. A common aid is the **scree plot**, the points (m, PVE_m), $m = 1, \ldots, p$, joined by segments: one takes the smallest M that explains a sizable part of the total variance, typically the M after which the PVE_m drop off and stay small (an **elbow** of the plot after the Mth component). There is no objective rule; the choice depends on the data and on the purpose.

5. Principal components of a random vector

Definition 22. Let $x = (x_1, \ldots, x_p)'$ be a random vector with $\Sigma = \mathrm{var}(x) \neq 0$. A **first principal component** of x is $a_1'x$, where the unit vector $a_1 \in \mathbb{R}^p$ maximizes $\mathrm{var}(a'x)$ over all unit vectors $a \in \mathbb{R}^p$. For $m \geqslant 2$, once $a_1, \ldots, a_{m-1}$ are chosen, an **mth principal component** is $a_m'x$, where the unit vector a_m maximizes $\mathrm{var}(a'x)$ over the unit vectors a for which $a'x$ is uncorrelated with each of $a_1'x, \ldots, a_{m-1}'x$.

Theorem 23. Let x and Σ be as in Definition 22, and $\Sigma = V\Lambda V'$ with $V = [v_1 \ \cdots \ v_p]$ orthogonal, $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_p)$, $\lambda_1 \geqslant \cdots \geqslant \lambda_p$ (as in Theorem 15).

(a) $\operatorname{var}(a'x) = a'\Sigma a$ and $\operatorname{cov}(a'x, b'x) = a'\Sigma b$ for $a, b \in \mathbb{R}^p$; and $\lambda_p \geqslant 0$.

(b) $v_1'x$ is a first principal component, with variance λ_1. Let $m \geqslant 2$ and $\lambda_{m-1} > 0$. If $v_1'x, \dots, v_{m-1}'x$ are the first $m-1$ principal components, then $v_m'x$ is an mth principal component, with variance λ_m.

(c) $v_1'x, \dots, v_p'x$ are pairwise uncorrelated, and the **total variance** $\operatorname{var}(x_1) + \cdots + \operatorname{var}(x_p) = \operatorname{tr}\Sigma = \lambda_1 + \cdots + \lambda_p$ is positive. So the first M principal components account for the proportion $(\lambda_1 + \cdots + \lambda_M)/\operatorname{tr}\Sigma$ of the total variance.

Proof. (a) Let $T \in \mathbb{R}^{2\times p}$ have the rows a', b', so $Tx = (a'x, b'x)'$ and $\operatorname{var}(Tx) = T\Sigma T'$ (Theorem 3(b)) has the entries $a'\Sigma a$, $a'\Sigma b$; $b'\Sigma a$, $b'\Sigma b$ (Lecture 1, Theorem 4(c),(d)); by Definition 1, its diagonal entries are $\operatorname{var}(a'x)$, $\operatorname{var}(b'x)$ and its $(1,2)$ entry is $\operatorname{cov}(a'x, b'x)$. Σ is PSD (Theorem 3(c)), so $\lambda_p \geqslant 0$ (Lecture 8, Theorem 14(a)).

(b)

- By (a), the first problem maximizes $a'\Sigma a$ over the unit a; for $a \neq 0$, $a'\Sigma a/a'a$ is the value at the unit vector $a/\|a\|$. By Theorem 15(a) the maximum is λ_1, at $a = v_1$.
- Let $m \geqslant 2$ and $\lambda_{m-1} > 0$; then $\lambda_l > 0$ for $l < m$. By (a) and $\Sigma v_l = \lambda_l v_l$, $\operatorname{cov}(a'x, v_l'x) = a'\Sigma v_l = \lambda_l a'v_l$, which is $0 \iff a \perp v_l$. So the mth problem maximizes $a'\Sigma a$ over the unit $a \perp v_1, \dots, v_{m-1}$.
- By Theorem 16, the maximum is λ_m, at $a = v_m$, and $\operatorname{var}(v_m'x) = v_m'\Sigma v_m = \lambda_m$.

(c) By (a), $\operatorname{cov}(v_m'x, v_l'x) = v_m'\Sigma v_l = \lambda_l v_m'v_l = 0$ for $l \neq m$. $\operatorname{var}(x_j) = \sigma_{jj}$ (Definition 1), so the total variance is $\operatorname{tr}\Sigma = \operatorname{tr}(V\Lambda V') = \operatorname{tr}(\Lambda V'V) = \operatorname{tr}\Lambda$ (Lecture 1, Theorem 10(b)), which is positive: every $\lambda_m \geqslant 0$ by (a), and $\Lambda \neq 0$, as $\Sigma = V\Lambda V' \neq 0$. □

Parts (b) and (c) are Theorem 18(c),(d) and Proposition 19 with Σ in place of S. In practice Σ is not known: n observations of x form the rows of a data matrix, and the principal components of the data (Definition 17), computed from S, take the place of those of x.

Example 24. Let $x = (x_1, \dots, x_p)'$ ($p \geqslant 2$) have $\operatorname{var}(x_i) = \sigma^2 > 0$ and $\operatorname{cov}(x_i, x_j) = \rho\sigma^2$ for $i \neq j$, where $0 < \rho < 1$: $\Sigma = \sigma^2\Big((1-\rho)I_p + \rho J_p\Big)$ ($J_p = \mathbf{1}_p\mathbf{1}_p'$). Find the principal components of x.

Solution.
$I_p = \bar{J}_p + C_p$, $J_p = p\bar{J}_p$; for $a \in \mathbb{R}^p$, $\bar{J}_p a = \bar{a}\mathbf{1}$, $C_p a = a - \bar{a}\mathbf{1}$ (Lecture 1, Definition 14, Example 16):

$$\Sigma = \lambda_1\bar{J}_p + \lambda_2 C_p, \qquad \lambda_1 = \sigma^2\Big(1 + (p-1)\rho\Big), \qquad \lambda_2 = \sigma^2(1-\rho), \qquad \lambda_1 - \lambda_2 = p\rho\sigma^2 > 0$$

$$\Sigma\mathbf{1} = \lambda_1\mathbf{1}, \qquad \Sigma a = \lambda_2 a \ \text{ if } \mathbf{1}'a = 0$$

$$\Sigma - \lambda_1 I_p = (\lambda_2 - \lambda_1)C_p, \qquad \text{so } \Sigma a = \lambda_1 a \iff C_p a = 0 \iff a = \bar{a}\mathbf{1}$$

$v_1 = \frac{1}{\sqrt{p}}\mathbf{1}$; $[v_1 \ \cdots \ v_p]$ is orthogonal for some $v_2, \dots, v_p$ (Lecture 6, Corollary 11(c)); then $\mathbf{1}'v_j = \sqrt{p}\, v_1'v_j = 0$, $\Sigma v_j = \lambda_2 v_j$ ($j \geqslant 2$): the eigenvalues are λ_1 and λ_2 ($p-1$ times) (Lecture 8, Theorem 9(b)); $\lambda_1 > \lambda_2 > 0$.
The principal components (Theorem 23):

- First: $v_1'x = \frac{1}{\sqrt{p}}(x_1 + \cdots + x_p)$, with variance λ_1. Unique up to sign: a unit a with $a'\Sigma a = \lambda_1$ has $\Sigma a = \lambda_1 a$ (Theorem 15(b)), so $a = \bar{a}\mathbf{1} = \pm v_1$.
- Second: $\operatorname{cov}(a'x, v_1'x) = \lambda_1 a'v_1$, so the constraint is $\mathbf{1}'a = 0$; every such unit a has $\Sigma a = \lambda_2 a$ and $\operatorname{var}(a'x) = \lambda_2$. Every $a'x$ with $\mathbf{1}'a = 0$, $\|a\| = 1$ is a second principal component: for $p \geqslant 3$ not unique, even up to sign (for $p = 2$ it is $\pm(x_1 - x_2)/\sqrt{2}$).
- The first accounts for $\lambda_1/\operatorname{tr}\Sigma = \Big(1 + (p-1)\rho\Big)/p$ of $\operatorname{tr}\Sigma = p\sigma^2$; for $p = 4$, $\rho = \frac{1}{2}$: $\frac{5}{8}$.

Practice. The Exercises and Problems of the textbook, Chapter 9.